Camera | Complete Illustrated Research Report

GEMSTAR Task 23: pixel decoding, color layout, pinhole sanity test, controlled fog images, ROI metrics, MATLAB and preprocessing adapters.

Evidence source: uploaded Task_23_Final_Research.zip. Figures and tables come from archived assets. This document explains established findings separately from assumptions and future validation.

Executive research conclusions

Archived Camera evidence supports RGB payload decoding, controlled static-scene images, perspective response to a 2 m cube displacement, five visual quality proxies under three weather settings, MATLAB luminance/contrast verification, and a validated software adapter. The results do not demonstrate image-based detection or tracked-object accuracy.

Weather	Center luminance	Center contrast	Center edge density	Center saturation
Clear	169.4138	45.9824	0.011366	38.7888
Fog100	178.8827	43.6135	0.010986	26.7923
Fog20	194.8957	31.0490	0.008970	12.6940

How each Camera subtask contributed

23.2-3A | Locate and decode Camera payload

Inspect OSI Camera payload field format, frame dimensions and byte count before interpreting pixel values; inspect plausible BGR, RGB and grayscale conversions.

23.3B/C | Layout hypotheses and verification

Compare color layout candidates and verify source pixel ordering. A raw 640x480x3 uint8 image contains 921,600 bytes, but correct channel order still matters for RGB/BGR conversion.

23.3D/E | Camera image quality and MATLAB

Compute brightness, contrast, edge metrics on selected frames, then reproduce supported quantities in MATLAB. Early sample images and metrics are not the later fully controlled fog study.

23.9 | Controlled perspective validation

Record baseline/displaced views of a 3 m cube, with front-face distances 8.5 and 10.5 m; measured widths 112 and 92 px. Source FoV and pinhole model predict 112.941 and 91.429 px, an exploratory sanity check, not calibrated reprojection accuracy.

23.12A/B | Weather and static calibration scene

Verify clear map, create fog MOR100/MOR20 conditions, verify runtime weather application and preserve original maps via backups/hashes.

23.12C.1/2 | Timestamp and matched image extraction

The first ten frames of each scenario were audited; the displayed comparison triplet is near 0.2 s (timestamps not identical). Images come from distinct recordings, not same-run synchronized sensors.

23.12C.5/6 | ROI feature extraction and CPU preprocessing

Extract central ROI x=128, y=96, w=384, h=288, as well as full image; compute mean luminance, contrast standard deviation, edge density, Laplacian variance and mean saturation; CPU code validates BGR shape, dtype and ROI.

23.12C.7/D | MATLAB, robustness and report

MATLAB agrees with Python central ROI luminance/contrast within the archived tolerance; eight invalid-input cases pass. Evidence manifest stores references; raw multi-GB runs are external.

23.14 | Intrinsic/mount audit

CameraSensor.cpp sets capture horizontal FoV from the configured value. Nominal K uses fx=fy=320 px, cx=320, cy=240 for 640x480, 90 degree horizontal FoV and square-pixel assumption; actual optical-frame reprojection still unverified.

23.15 | Camera adapter integration

Wrap the validated CPU preprocessing function, preserve image features and metadata, match reference grayscale/edges/features on a synthetic BGR image (7/7 checks). No recorded-OSI end-to-end Camera adapter run yet.

Matched-time ROI measurement and uncertainty

The controlled Camera comparison images correspond to Clear frame 2 at 0.189942 s, Fog100 frame 2 at 0.203152 s, and Fog20 frame 2 at 0.192330 s. These are near-time images of a static validation arena, not pixel-perfect matched frames. Fog20 showed higher mean luminance, reduced contrast and lower saturation in the measured ROI. These are appearance features, not calibrated sensor confidence.

Weather	Full luminance	Full contrast	Full Laplacian var	Full saturation
Clear	125.5095	73.9714	317.6918	80.0809
Fog100	139.5759	70.5069	275.8732	35.4572
Fog20	165.4152	56.8249	142.0111	16.0378

What the camera intrinsic matrix does and does not prove

The cube width comparison supports nominal focal length scale from the horizontal FoV. It does not verify Camera optical-axis conversion, effective mounted pose, object-specific correspondence, or pixel reprojection residuals of LiDAR/Radar points. The standalone calibration camera mount differs from the later vehicle-mounted weather Camera setup.

Weather-image evidence from extracted Camera frames

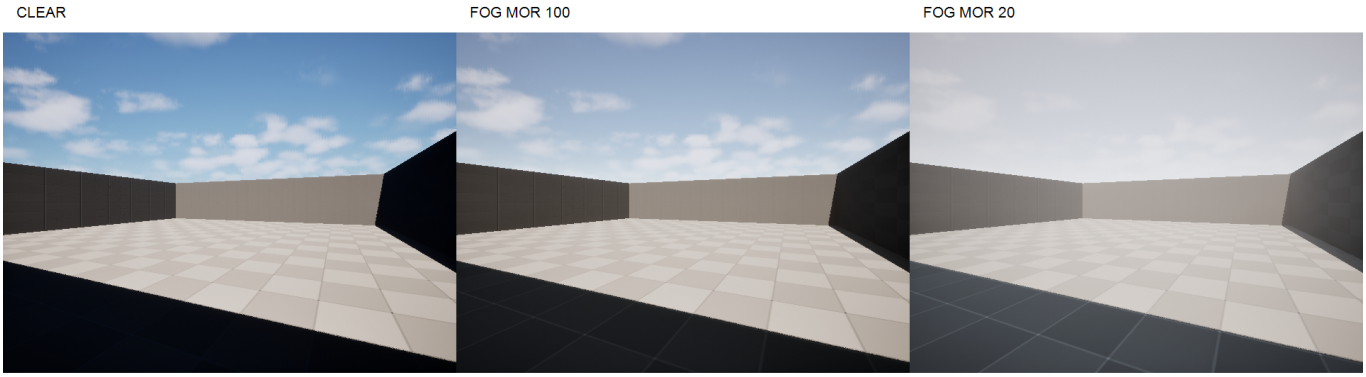


Figure 1. task23_12C_03_three_weather_comparison.png - Direct archived three-condition image panel. Interpret visibility changes in the static scene; no detection targets were verified.

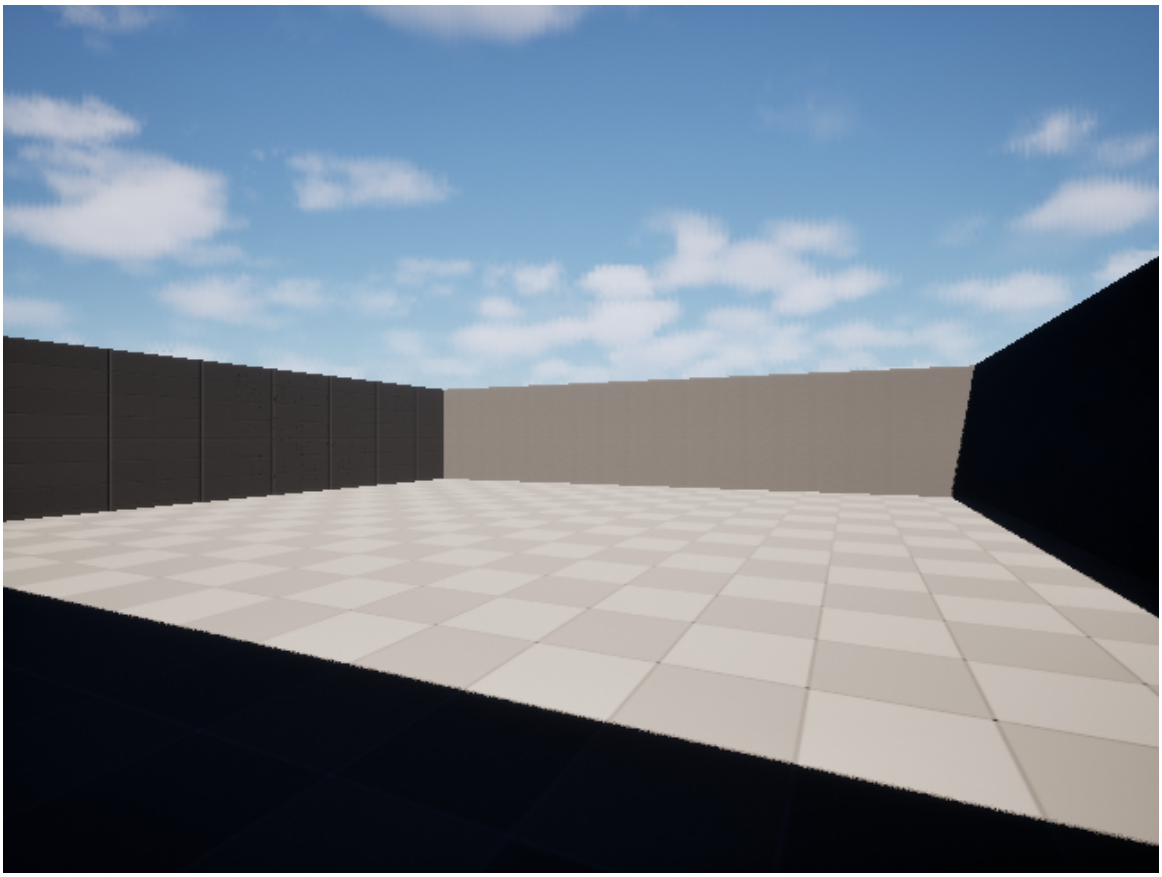


Figure 2. CLEAR_t0200ms.png - Clear reference near 0.2 s, extracted from its own recording.

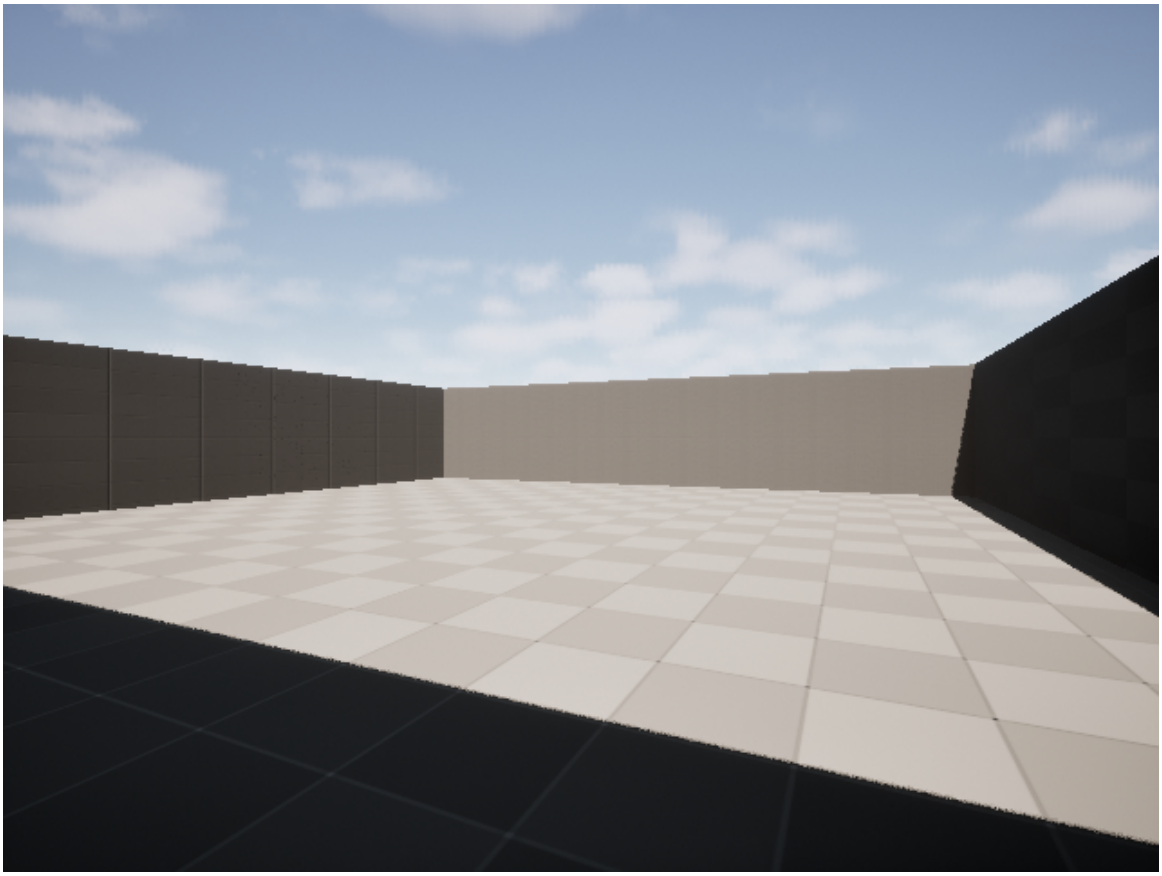


Figure 3. FOG100_t0200ms.png - Fog MOR100 near 0.2 s, extracted from a separate recording.

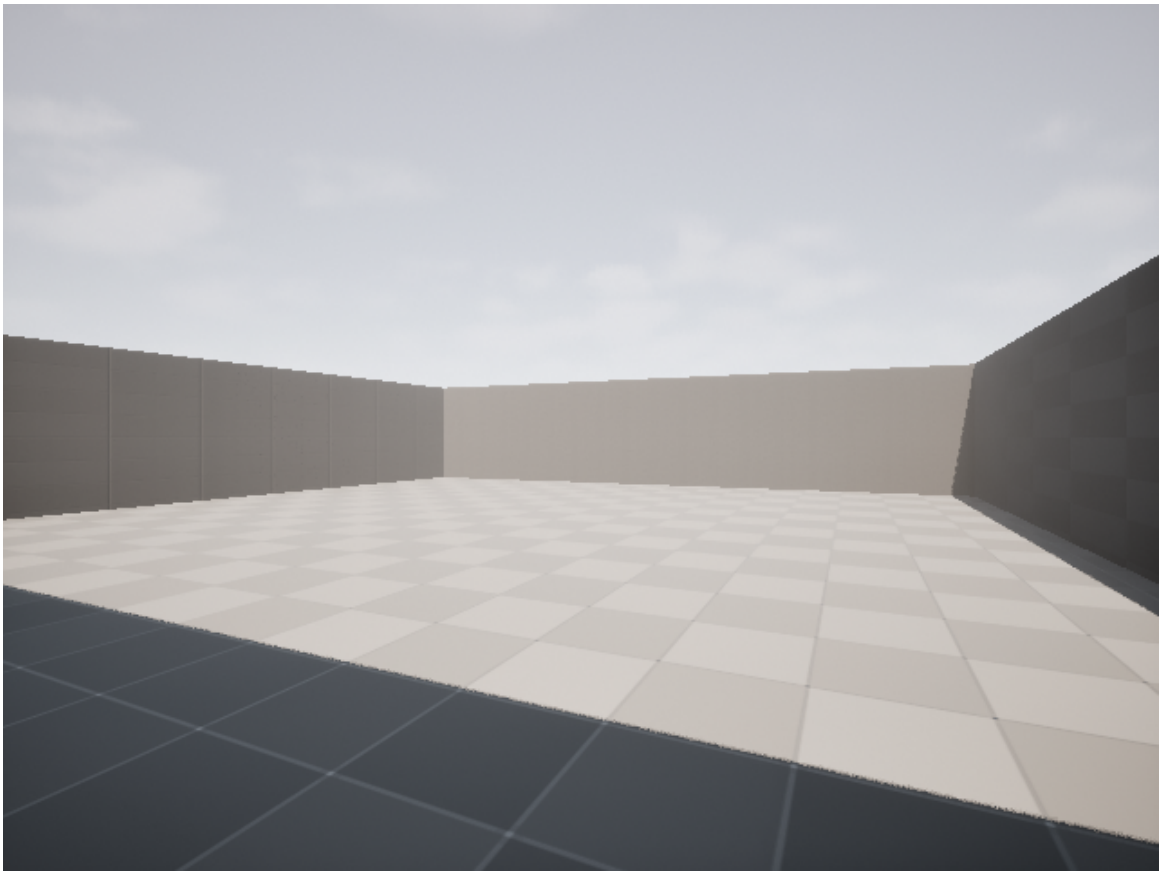


Figure 4. FOG20_t0200ms.png - Fog MOR20 near 0.2 s, extracted from a separate recording.

Illustrated figure-by-figure interpretation

Every figure below is embedded from the supplied archive. Captions explain the research question, what the figure can support, and any key interpretation limit. Images are evidence from historical subtasks, not independently recreated measurements.

Task 23.3 - Camera layout and MATLAB diagnostics

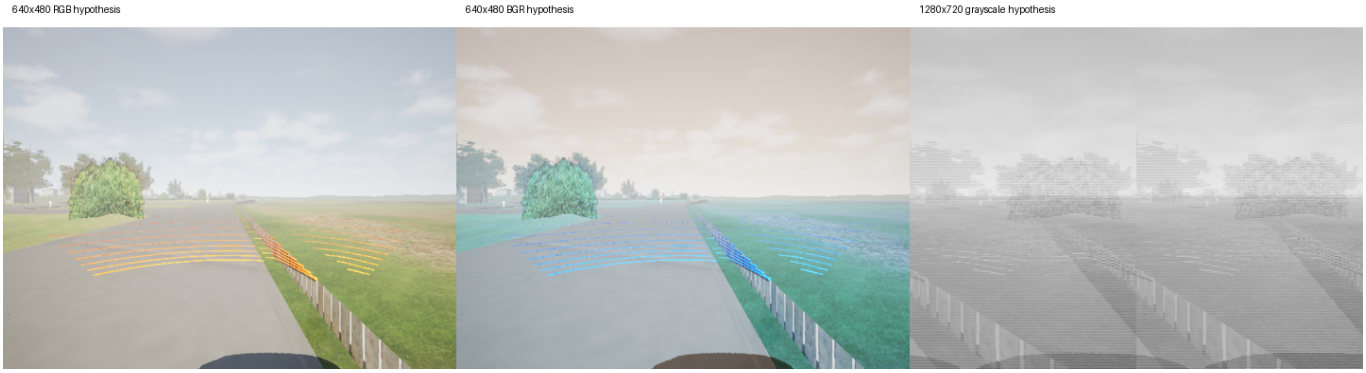


Figure 5. task23_03b_pair1_camera_layout_comparison.png - Compared alternative interpretations of the raw Camera payload; supports the archived RGB layout choice only when paired with the format-validation CSV, not object detection.

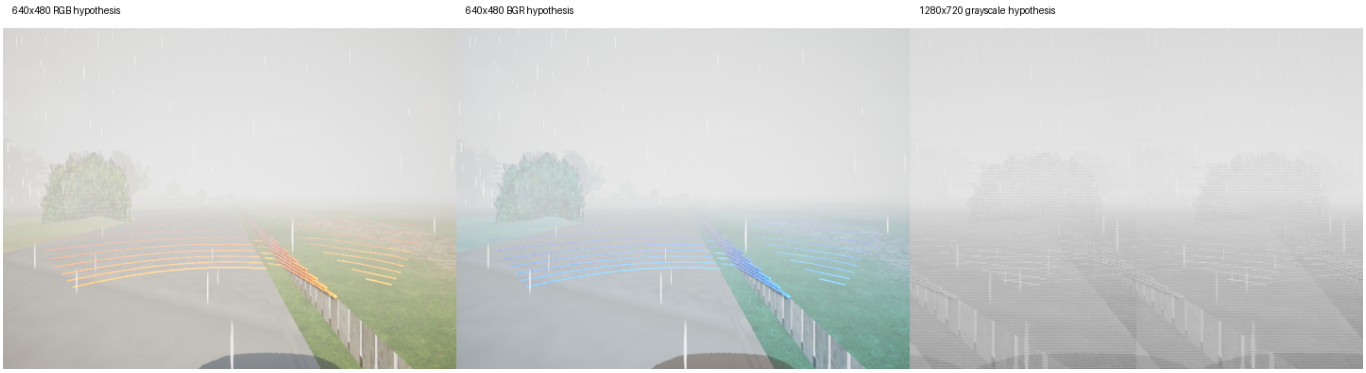


Figure 6. task23_03b_pair2_camera_layout_comparison.png - Compared alternative interpretations of the raw Camera payload; supports the archived RGB layout choice only when paired with the format-validation CSV, not object detection.

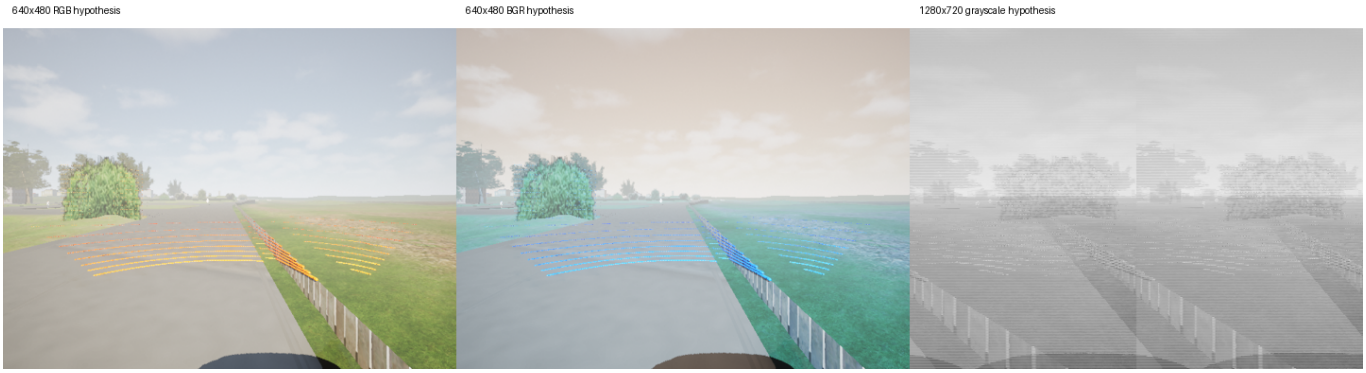


Figure 7. task23_03b_pair3_camera_layout_comparison.png - Compared alternative interpretations of the raw Camera payload; supports the archived RGB layout choice only when paired with the format-validation CSV, not object detection.

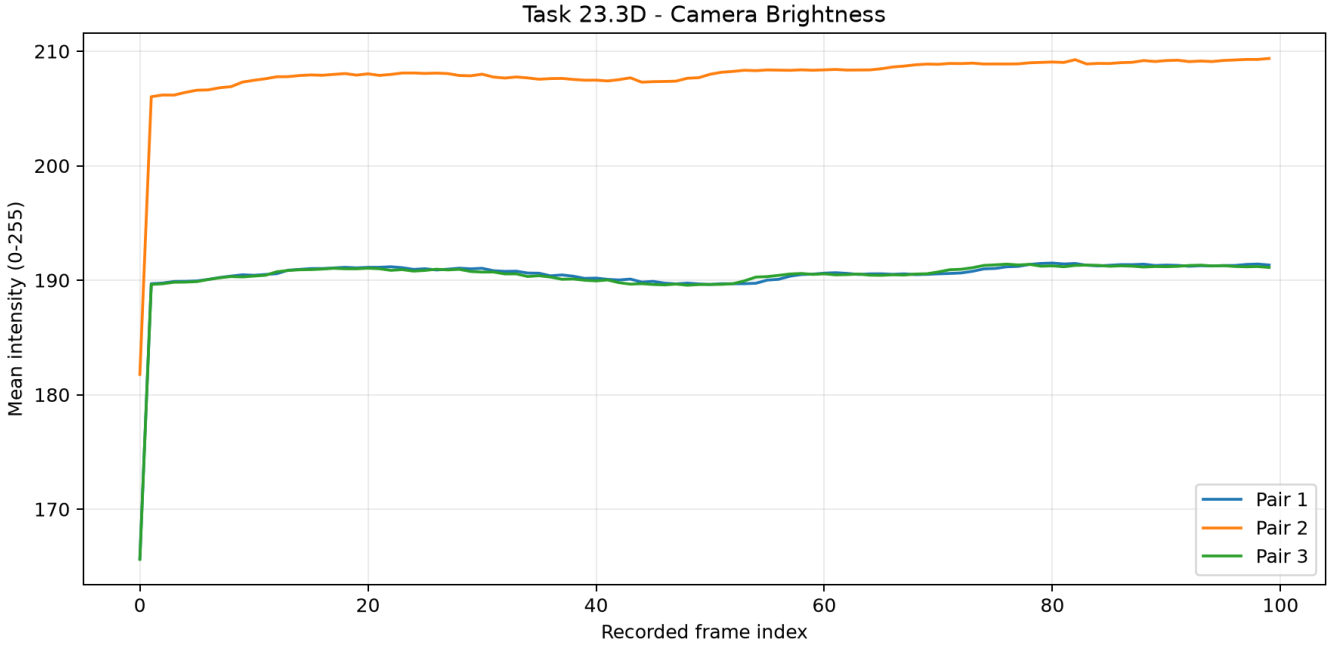


Figure 8. task23_03d_camera_brightness.png - Tracks frame luminance extracted from Camera payloads; changes describe image appearance, not detection confidence.

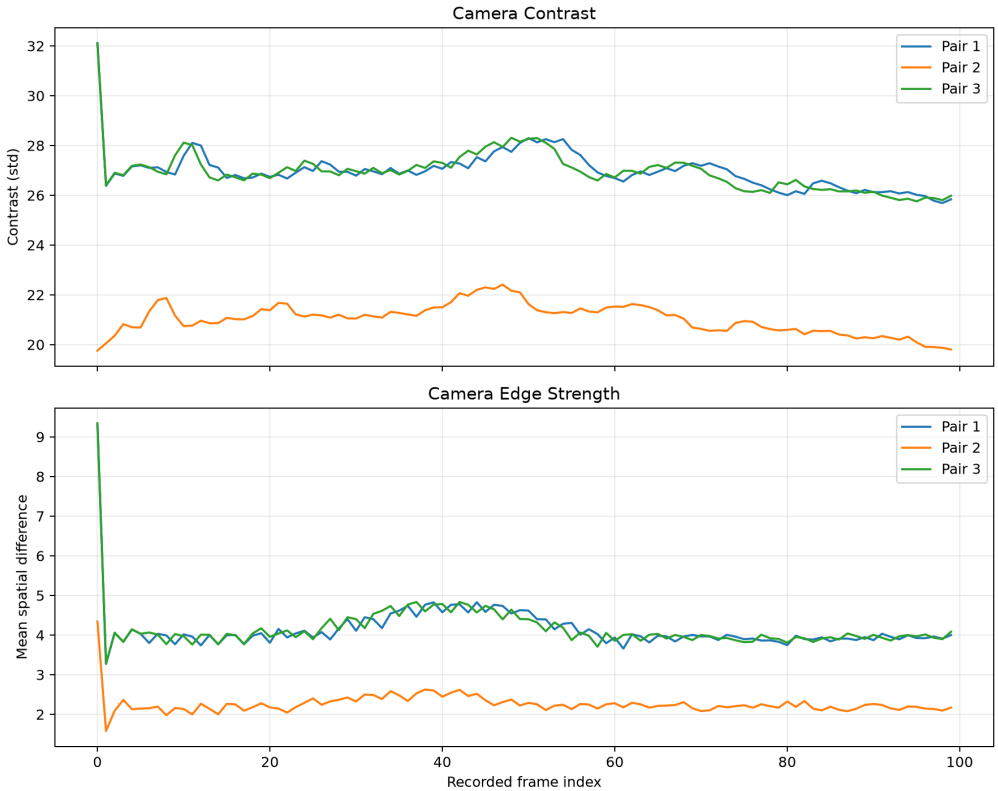


Figure 9. task23_03d_camera_contrast_edges.png - Displays image contrast and edge proxies; these quantify texture response without establishing classification or tracking accuracy.

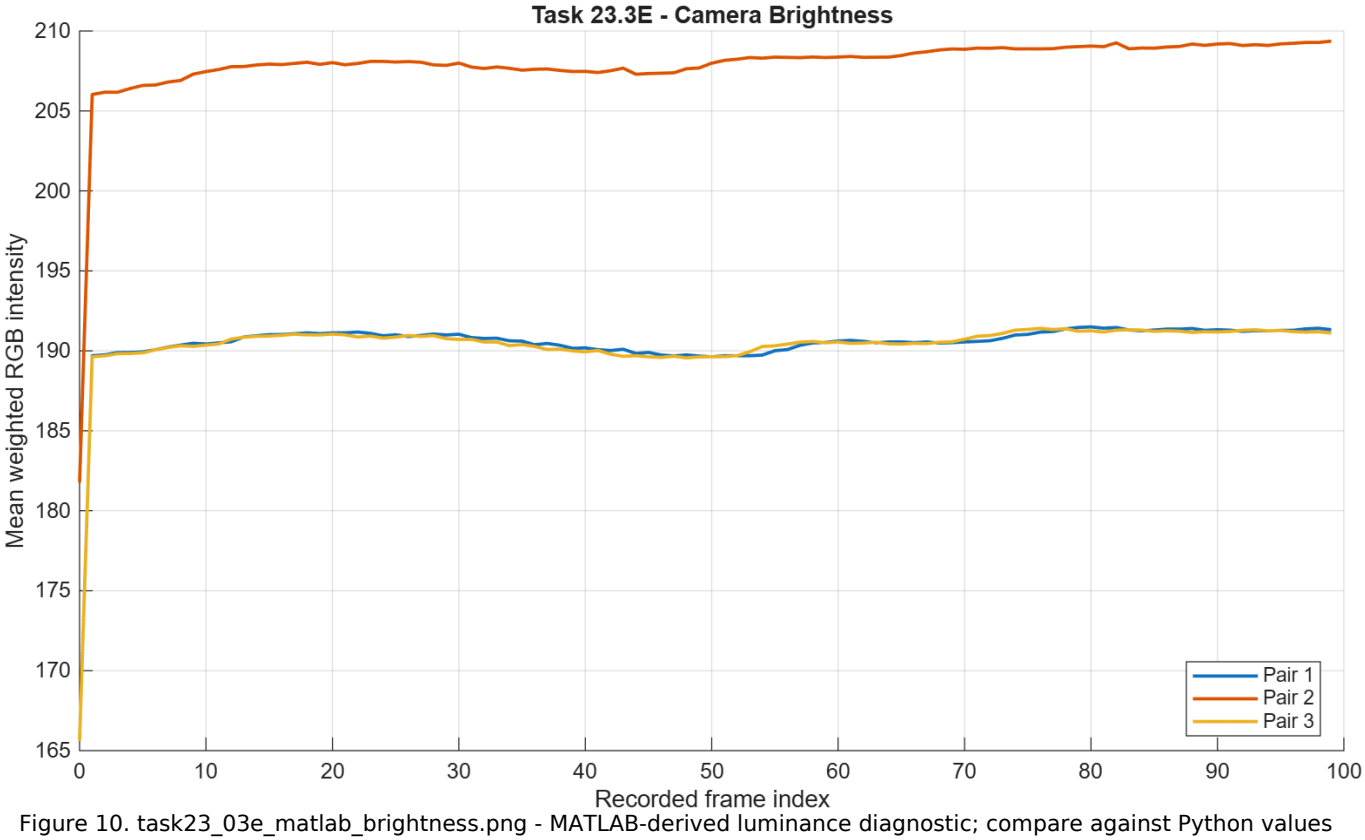


Figure 10. task23_03e_matlab_brightness.png - MATLAB-derived luminance diagnostic; compare against Python values in the same dataset, not between unrelated runs.

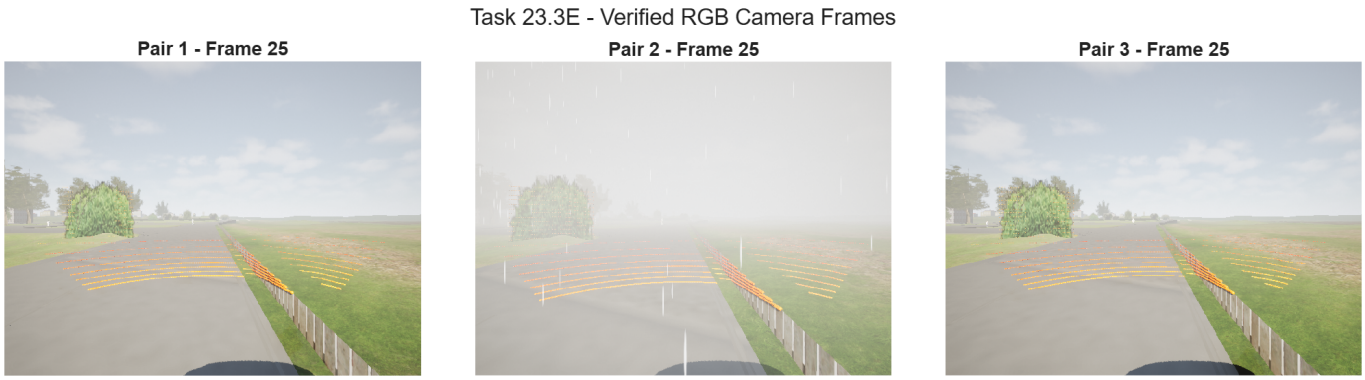


Figure 11. task23_03e_matlab_camera_comparison.png - MATLAB and Python Camera-analysis comparison or image summary; illustrates cross-tool interpretation, with metric-specific agreement established by the associated CSV.

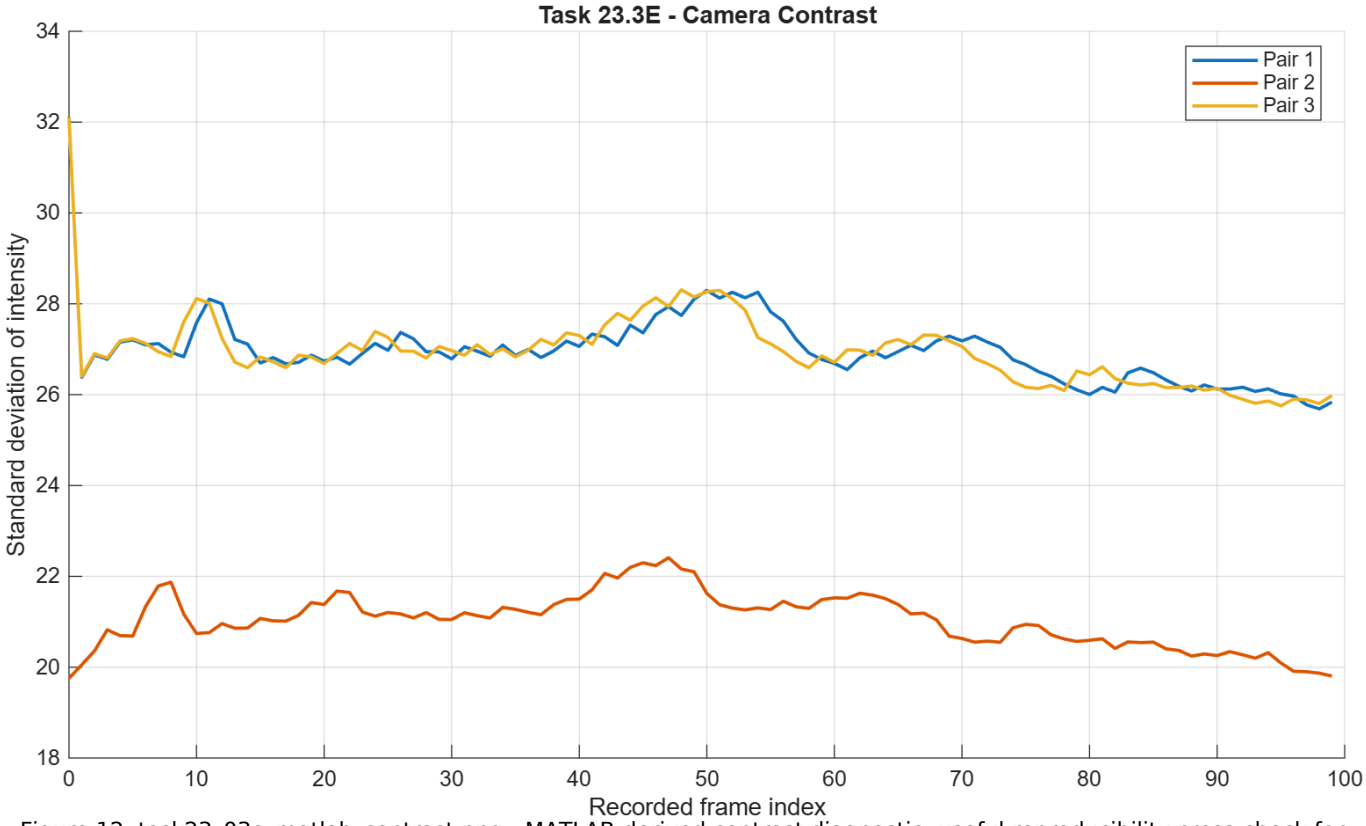


Figure 12. task23_03e_matlab_contrast.png - MATLAB-derived contrast diagnostic; useful reproducibility cross-check for image statistics.

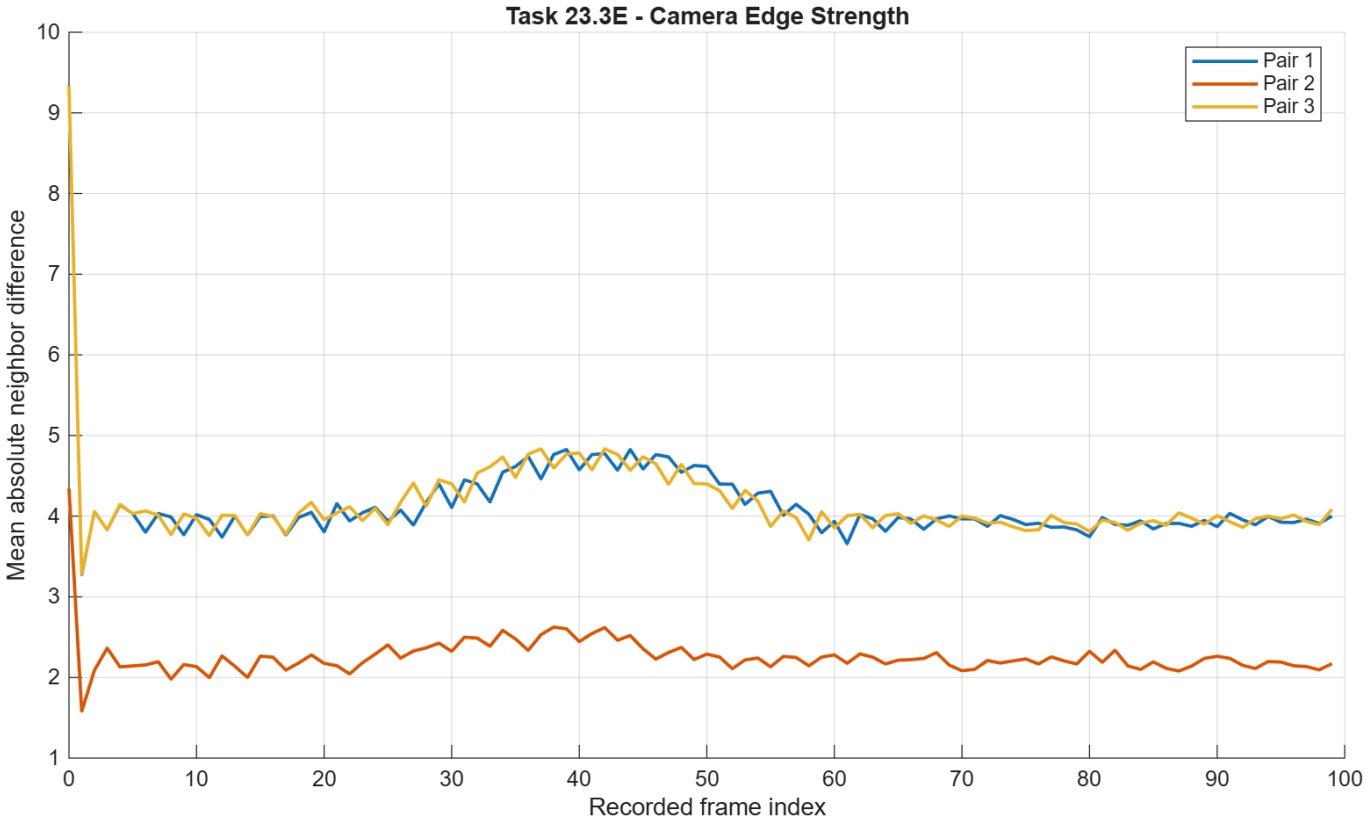


Figure 13. task23_03e_matlab_edge_strength.png - MATLAB edge-strength diagnostic; exploratory edge proxy, not externally calibrated performance.

Task 23.9 - geometric calibration and target displacement

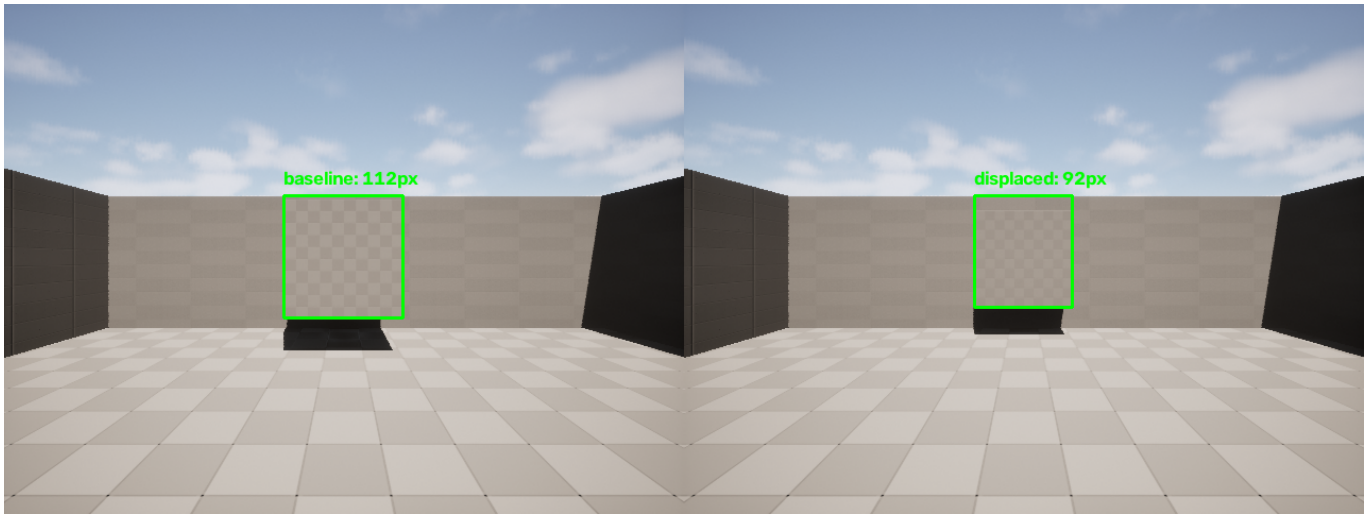


Figure 14. task23_09_auto_detected_cubes.png - Automatic image bounding boxes for cube target at two positions; edge detection and clipping can bias pixel widths.

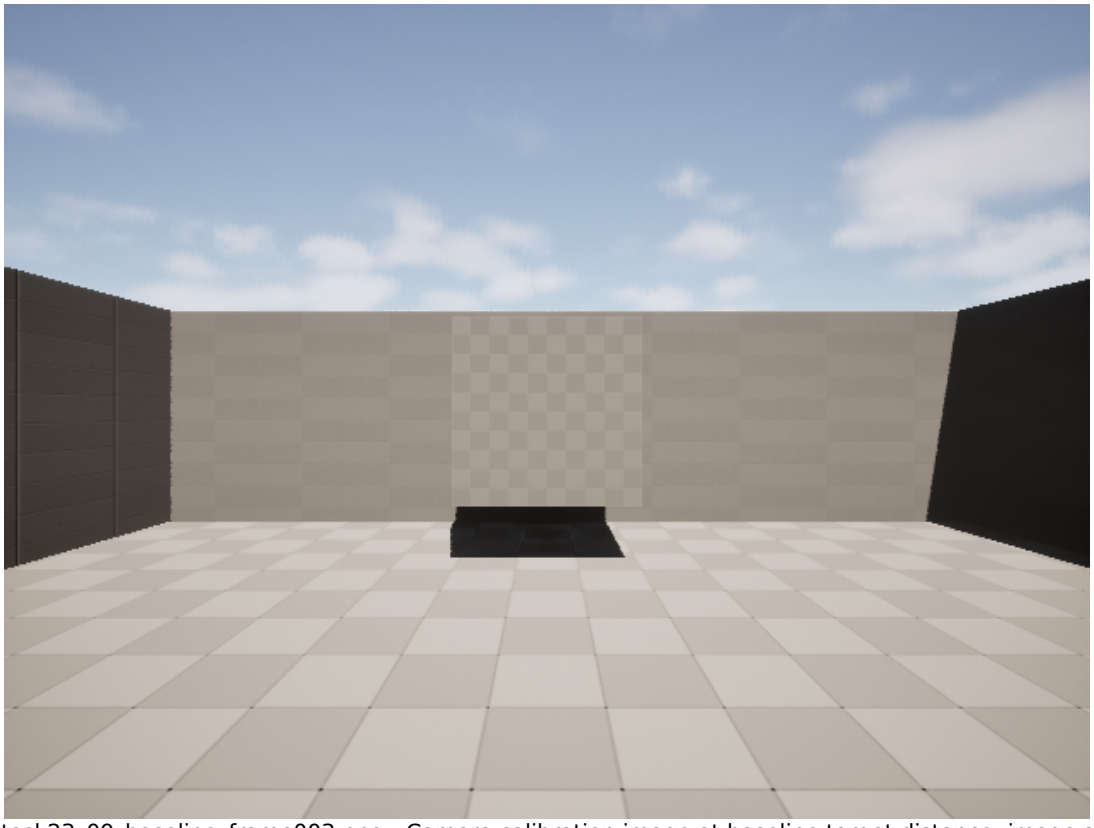


Figure 15. task23_09_baseline_frame003.png - Camera calibration image at baseline target distance; image serves as source evidence for measured pixel width.

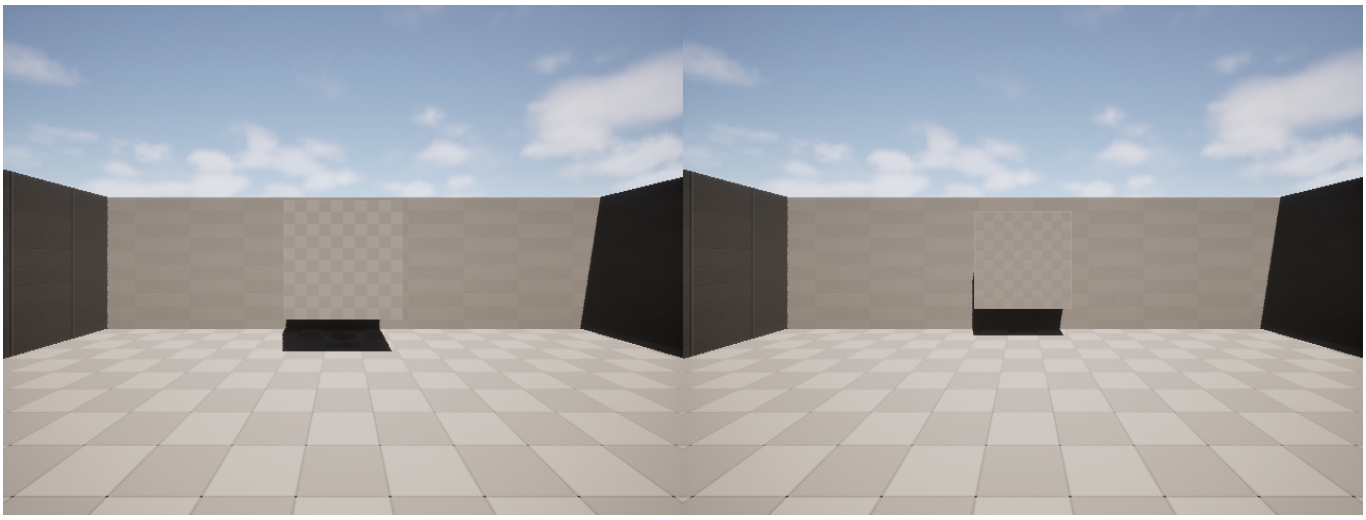


Figure 16. task23_09_baseline_vs_displaced.png - Camera target comparison after 2 m change in distance; apparent width change agrees approximately with perspective geometry.

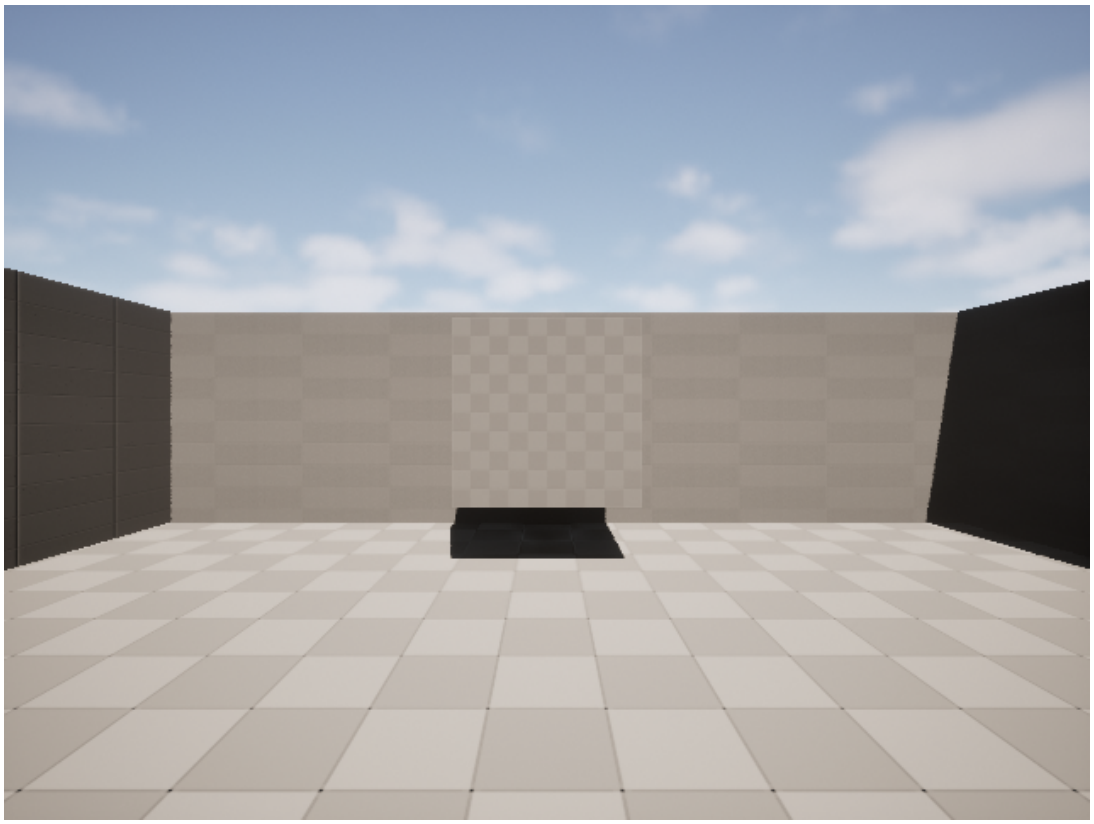


Figure 17. task23_09_camera_baseline_frame001.png - Camera calibration image at baseline target distance; image serves as source evidence for measured pixel width.

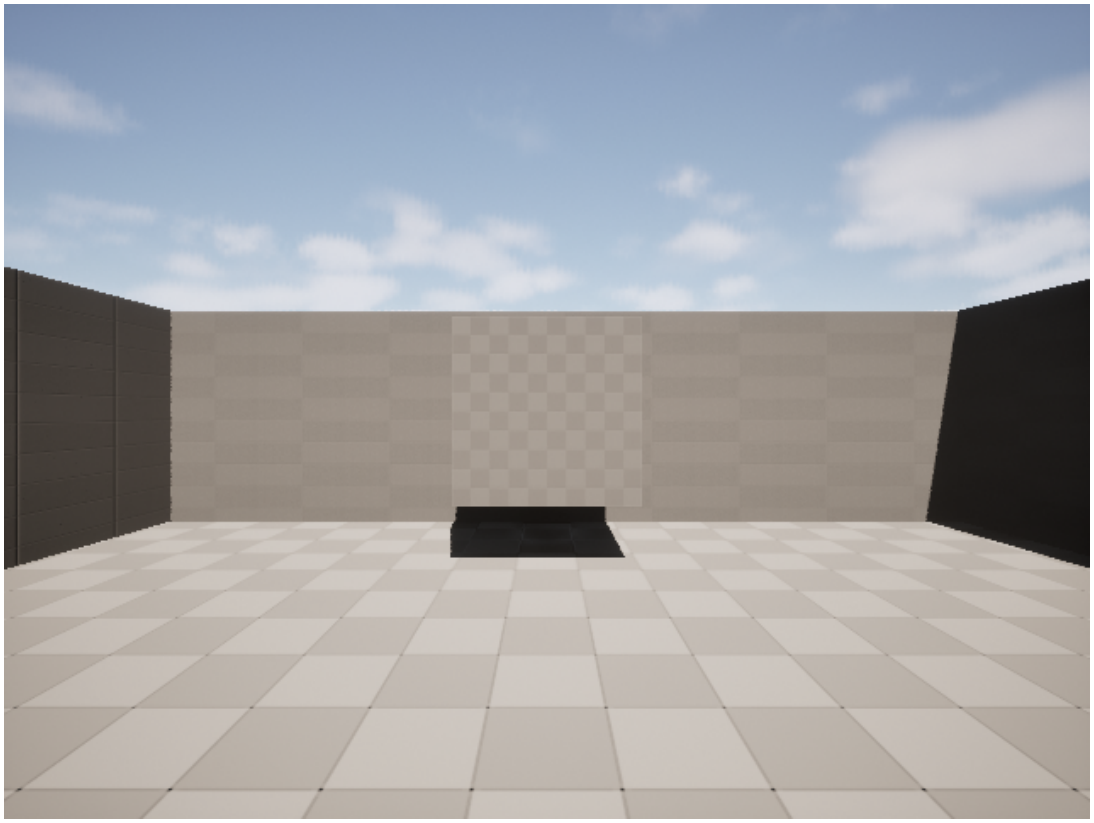


Figure 18. task23_09_camera_baseline_frame003.png - Camera calibration image at baseline target distance; image serves as source evidence for measured pixel width.

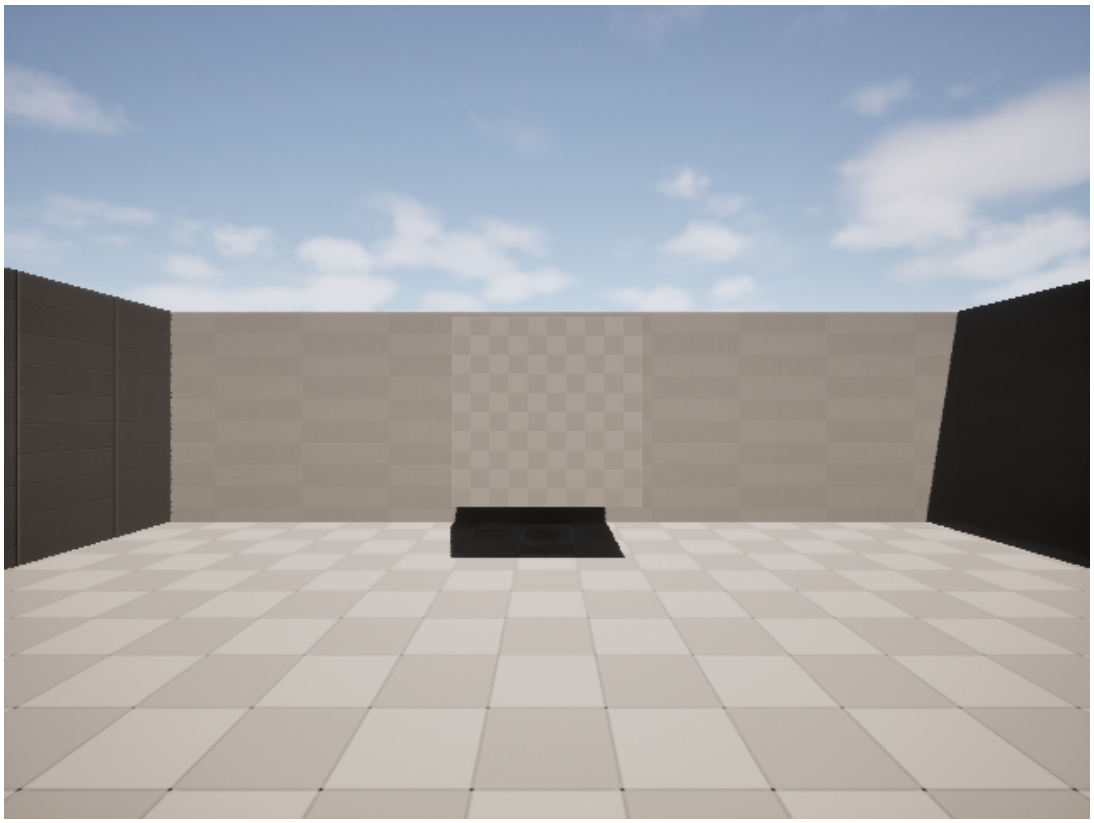


Figure 19. task23_09_camera_baseline_frame006.png - Camera calibration image at baseline target distance; image serves as source evidence for measured pixel width.

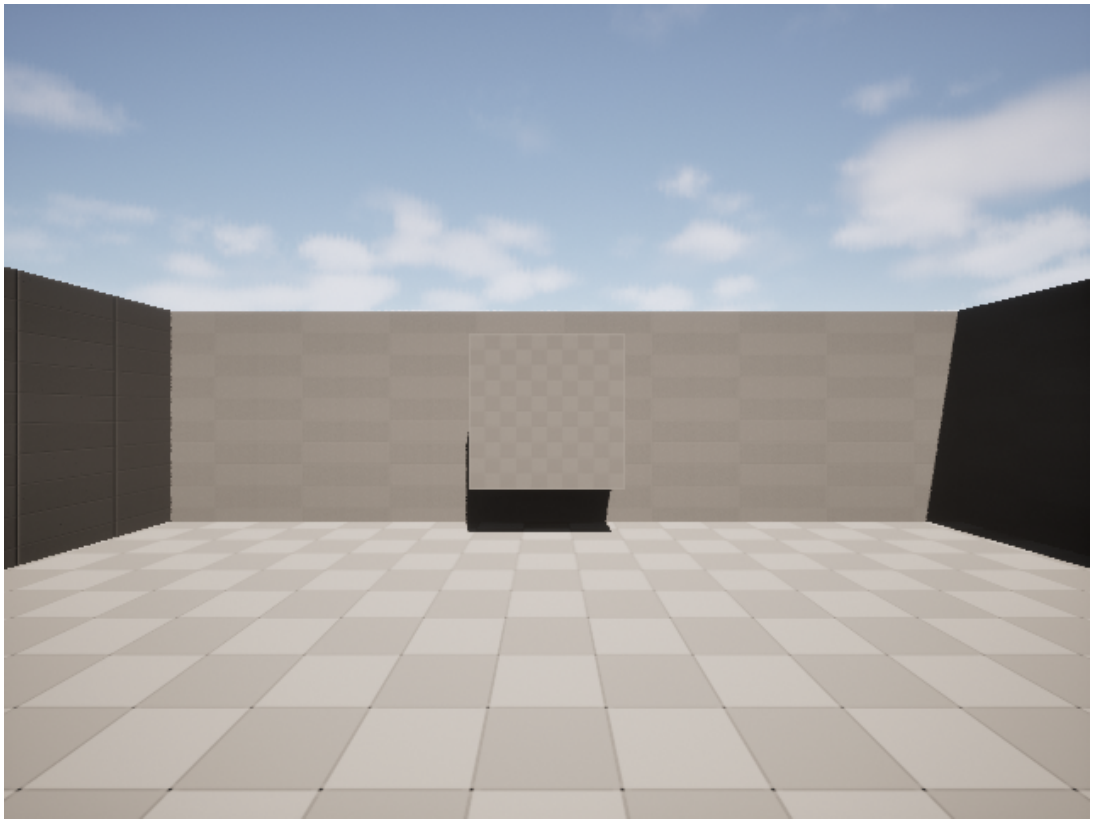


Figure 20. task23_09_displaced_frame003.png - Camera calibration image with displaced cube; compare using the same bounding-box method.

Task 23.12 - Python/MATLAB weather comparison

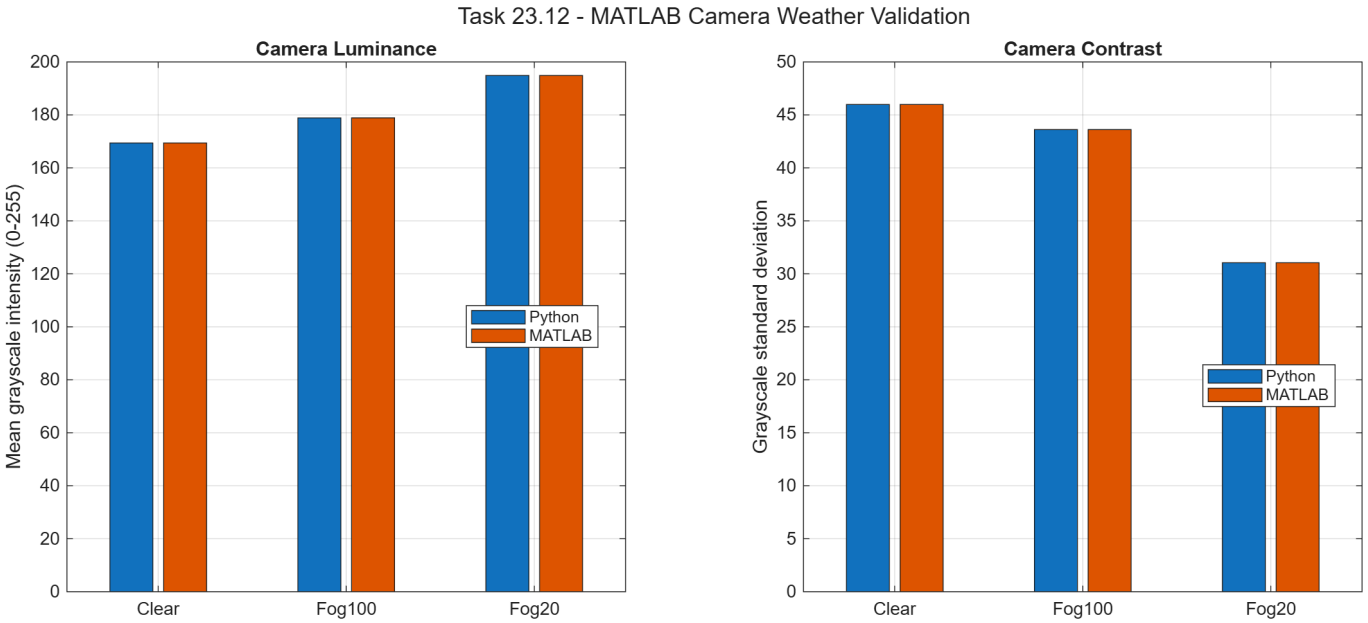


Figure 21. task23_12C_07_matlab_camera_weather_comparison.png - MATLAB validation of selected Camera central ROI luminance and contrast under three controlled weather settings.

Detailed subtask and code inventory

The table below is organized by archived script name and research objective. A filename identifies work that exists; it is not by itself a claim that every planned experiment or verification was completed.

Task 03a - source scripts (1)

task23_03a_camera_format_inspection.py - Inspect recorded or simulated payloads before accepting analysis.

Task 03b - source scripts (1)

task23_03b_camera_decode_candidates.py - Interpret encoded Camera/Radar/OSI measurements.

Task 03c - source scripts (1)

task23_03c_verify_color_format.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 03d - source scripts (1)

task23_03d_camera_quality_analysis.py - Calculate image/sensor-quality metrics rather than detection accuracy.

Task 03e - source scripts (1)

task23_03e_matlab_camera_validation.m - MATLAB cross-tool calculation/plot based on exported research data.

Task 09 - source scripts (9)

task23_09_auto_cube_measure.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_09_camera_set_map.py - Modify or prepare experimental map (historical operation; do not rerun).

task23_09_camera_verify_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_09_compare_camera_images.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_09_decode_camera_frames.py - Interpret encoded Camera/Radar/OSI measurements.

task23_09_displace_camera_target.py - Controlled change to target geometry for calibration.

task23_09_finalize_camera.py - Consolidate evidence and create a research-ready result.

task23_09_measure_camera_cube.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_09_verify_camera_displacement.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 12A - source scripts (4)

task23_12A_04C_camera_weather_payload_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_12A_set_camera_clear_map.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_12A_verify_camera_clear_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_12A_verify_clear_map_evidence.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 12B - source scripts (5)

task23_12B_set_camera_fog100_map.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_12B_set_camera_fog20_map.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

task23_12B_verify_clear_camera_payload.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_12B_verify_fog100_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

task23_12B_verify_fog20_map.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 12C - source scripts (7)

task23_12C_01_camera_timing_audit.py - Inspect source, data structure, indexing, or evidence assumptions.

task23_12C_02_extract_matched_camera_images.py - Extract selected frames, sensor values, or metrics.

task23_12C_05_camera_quality_metrics.py - Calculate image/sensor-quality metrics rather than detection accuracy.

task23_12C_06_camera_cpu_preprocessing.py - Implement sensor-specific preprocessing baseline.

task23_12C_07_matlab_camera_validation.m - MATLAB cross-tool calculation/plot based on exported research data.

task23_12C_07_run_matlab_and_plot.m - MATLAB cross-tool calculation/plot based on exported research data.

task23_12C_07_run_matlab_and_plot_original.m - MATLAB cross-tool calculation/plot based on exported research data.

Task 12D - source scripts (2)

task23_12D_01_camera_robustness_tests.py - Run defined behavior or invalid-input checks.

task23_12D_02B_verify_before_restore.py - Read-only verification of scenario artifacts, data validity, or configuration.

Task 14B - source scripts (1)

task23_14B_03F_camera_intrinsic_sanity_check.py - Supporting research experiment or data-preparation script. Consult original source for exact operations.

Task 15 - source scripts (1)

task23_15_camera_adapter.py - Wrap validated sensor-specific operation into common interface.

CSV evidence and reproducibility register

CSV file	Rows	Example fields
task23_03a_camera_format_candidates.csv	9	PairID, Frame, ImageBytes...
task23_03b_camera_decode_summary.csv	3	PairID, Frame, ImageBytes...
task23_03c_verified_camera_format.csv	3	PairID, Frame, ChannelFormatValues...
task23_03d_camera_frame_metrics.csv	300	PairID, Frame, Width...
task23_03d_camera_recording_summary.csv	3	PairID, Frames, BrightnessMean_Mean...
task23_03e_matlab_cross_validation.csv	9	PairID, Frame, BrightnessError...
task23_09_camera_auto_measurements.csv	2	condition, left, right...
task23_09_camera_baseline_displaced.csv	2	condition, cube_front_face_m, frame...
task23_09_camera_cube_measurements.csv	2	condition, distance_m, x_min...
task23_09_camera_frame_inspection.csv	6	frame, image_bytes, unique_sample_colors...
task23_12A_04C_camera_weather_payload_audit.csv	15	Weather, SourceFile, Frame...
task23_12C_01_camera_timing_audit.csv	30	scenario, frame, timestamp_seconds...
task23_12C_02_matched_camera_frames.csv	3	scenario, frame_index, timestamp_seconds...

CSV file	Rows	Example fields
task23_12C_05_camera_quality_metrics.csv	6	scenario, region, mean_luminance...
task23_12C_07_matlab_camera_validation.csv	3	Scenario, MATLAB_Luminance, Python_Luminance...
task23_12D_01_camera_robustness_tests.csv	8	test, expected_accept, actual_accept...
task23_14B_03F_camera_intrinsic_sanity_check.csv	2	Condition, DistanceM, CubeFaceWidthM...

Missing figures, why they matter, and how to create them

Figure / experiment	Why it is needed	Source / feasibility
Annotated central ROI overlay	Readers need to know where 384x288 center crop comes from	Can generate now using archived clear/fog PNGs; fixed x/y/w/h from report.
Five normalized weather-feature panels	One accessible comparison of all five Camera features, with units and separate axes	Can generate now from task23_12C_05_camera_quality_metrics.csv.
Multi-frame metric distribution	Single representative triplet cannot establish variation/CI	Requires measuring representative distinct frames from raw OSI / extracted images.
Camera-LiDAR and Camera-Radar pixel reprojection residuals	Critical extrinsic and optical-axis validation	Matched same-run objects and ground-truth correspondences; do before Task 24 accuracy claims.

Scientific interpretation and handover

The checked results are a reproducible software and simulator preprocessing baseline. They do not establish real-world accuracy, dynamic multi-object perception, CPU-versus-FPGA energy improvement, or measured deadlines. Those claims require separate matched scenarios, explicit ground-truth evaluation, and physical hardware benchmarking.

All embedded plots/images are original archived figures except where a figure is explicitly described as a newly synthesized summary. The archived source file name accompanies each caption. Raw multi-gigabyte OSI sources are not embedded and were not reprocessed for these PDFs.